TEXORA · LECTURE SHEET

Tappet Shedding

Shedding by rotating cams: how the profile of a tappet — its lift portion and its dwell portion — is turned through a follower, lever and treadle into the rise and fall of a heald frame, and why this simplest of all shedding systems is also the most limited.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 4 · LEVEL 2 TERM 1

1. The tappet principle

A **tappet** is a cam: a disc or plate whose edge is not a true circle but a deliberate profile, machined so that its radius varies around its circumference in a planned way. As the tappet rotates at constant speed, a follower resting on its edge is pushed out where the radius grows, held still where the radius is constant, and allowed to return where the radius falls. Rotation in, reciprocating lift out — that conversion is the whole principle, and shedding is one of its oldest textile uses.

There is, in principle, **one tappet for each heald frame**, all mounted on a common shaft and keyed at the angles the pattern requires, so each frame's position at any instant is fixed by the profile under its follower. The pattern is stored in metal: to change the weave, the tappets must be changed. Once set, the system repeats exactly with no reading error — its strength, and its weakness (Section 6).

The motion path from cam to warp runs in a chain, and every link must be understood:

1. The rotating **tappet** presents its varying radius to the follower.
2. The **follower** — usually a roller at the end of a follower arm — rides on the profile and converts radius change into movement, rolling rather than sliding so the profile is not worn away.
3. A **lever system** (often a treadle lever worked from below, with connecting rods or **lamellae — flat links, often called lamia**) transmits the movement to the heald frame, multiplying or reducing it: the lever-arm ratio decides how much heald lift 1 mm of throw produces.
4. The **heald frame** rises or falls with all its ends, forming one layer of the shed; the other tappets position their frames at the same instant to form the other layer.

Return motion is provided by springs or a reversing roller motion (**negative** system, Section 4) or positively by the tappet arrangement itself (**positive** system). The tappet shaft is geared rigidly to the crankshaft, so the pattern cannot drift out of time unless a gear or key fails.

2. Tappet construction — lift and dwell in the profile

Look at a single tappet's outline and divide it, in the direction of rotation, into two kinds of arc:

- **The lift (and lowering) portion.** Radius changes steadily between minimum and maximum; while this arc passes, the follower and heald frame travel — rising as radius grows, falling as it shrinks. Its angular length, with loom speed, fixes how quickly the shed opens and closes; too steep a rise jerks the frame and the warp.
- **The dwell portion.** Radius is constant — maximum for a frame staying fully up, minimum for one staying fully down. While it passes, the follower does not move: the heald is held at full lift (or fully lowered) and the shed held fully open, giving the shuttle a stationary shed to cross (Topic 3, Section 6).

Maximum minus minimum radius is the **throw**. Throw is the displacement at the follower; lift at the heald eye is throw × lever ratio of the linkage. Both figures are needed to specify or check a tappet — confusing throw with lift is a common setting error.

Tappets are of cast iron or steel, with the working edge hardened. The profile must be smooth: any flat or worn step is copied into the heald motion as a hesitation in the shed, repeated every revolution.

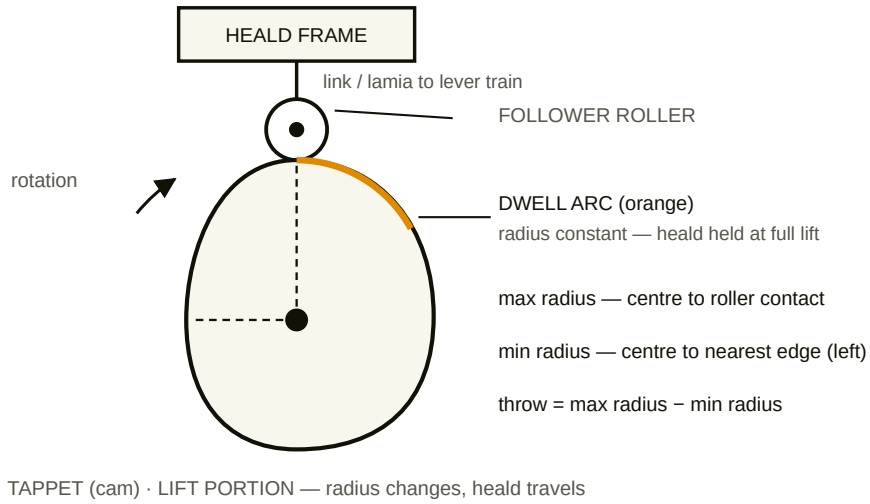


Figure 1 — A tappet and its drive to one heald frame. On the dwell arc the radius is constant and the heald is held still at full lift; on the lift portion the radius changes and the heald travels. The throw (maximum minus minimum radius) is multiplied by the lever ratio to give the lift at the heald eye.

3. The drive — the under-shaft at half crank speed

On the conventional loom the tappets sit on the **bottom shaft (under-shaft)**, geared from the crankshaft. For **plain weave** the pattern repeats every two picks — frame A up / B down, then the reverse — so a tappet must complete one revolution in two picks. The bottom shaft therefore turns at **half crankshaft speed**: one tappet revolution = two crank revolutions = two picks.

$$\text{bottom-shaft speed} = \text{crankshaft speed} \div 2 \quad \text{(for a 2-pick plain-weave repeat)}$$

The rule generalises: one tappet revolution covers exactly one complete pick repeat. A four-pick repeat needs the tappets to turn once in four picks — a quarter of crank speed. Getting this wrong produces a different weave altogether, because frames change places on the wrong picks; check it first when a loom's pattern does not match its tappets.



Figure 2 — The two-tappet plain-weave arrangement. The two cams are set half a revolution apart, so the frames exchange places on every pick and the pair completes its cycle in two picks — one revolution of the half-speed shaft.

4. Negative and positive tappets

FEATURE	NEGATIVE TAPPET	POSITIVE TAPPET
How a frame is lifted	By the tappet, through the follower and treadle linkage.	By the tappet arrangement.
How a frame is lowered (returned)	**Not** by the tappet. Springs, dead weights or a top reversing-roller motion pull the frame back down as the tappet radius falls and *allows* it to descend.	By the mechanism itself — for example paired cams or a cam-and-yoke arrangement that drives the follower in both directions, so the frame is positively controlled up and down.
Control of the shed	Lift is positive, return only indirect; at high speed the return can lag, giving an uneven shed.	Both layers placed positively every pick; shed position exact.
Complexity and cost	Simpler and cheaper; widely used for simple weaves at moderate speeds.	More parts and precise setting; justified by speed, heavy warp or exactness.
Typical consequence of wear	Weak springs → frames fail to follow the falling profile → floats, slack ends and an unclear shed.	Wear shows as lost motion and timing error rather than failure to return.

Most elementary descriptions describe the negative system — tappets lift, springs return. State for any laboratory loom which system is fitted; fault-finding (Section 7) differs for the two.

5. Dwell — how long the shed stays open

Dwell is the portion of the profile over which the radius is constant at its maximum, so the heald is held at full lift and the shed fully open. A common design figure is dwell for **about one-third of the pick cycle** — in pick-cycle terms: of the 360° of crank rotation per pick, roughly 120° passes while the shed is held fully open. Do not mix this with angles on the tappet itself: the tappet turns more slowly than the crank (half speed for plain weave), so tappet angles and crank angles differ by that gear ratio. The worked example below is done entirely in pick-cycle terms.

WORKED EXAMPLE 1 — HEALD LIFT FROM TAPPET THROW

A tappet has a maximum radius of 46 mm and a minimum radius of 32 mm. Its follower drives the heald frame through a lever system with a ratio of 1.6 (heald movement = 1.6 × follower movement). Find the heald lift.

Throw = maximum radius − minimum radius = 46 − 32 = **14 mm**.

Heald lift = throw × lever ratio = 14 × 1.6 = **22.4 mm**.

That 22.4 mm is the travel of the heald eye. Whether it gives a clear shed is then judged by the geometry of Topic 3: the height available at the shuttle is always less than the lift at the eye.

WORKED EXAMPLE 2 — DWELL TIME

A tappet is designed so that the heald dwells fully raised for 1/3 of the pick cycle. The loom runs at 200 picks per minute. How long, in seconds, is the shed held fully open on each pick?

Time for one pick = 60 s ÷ 200 picks = **0.30 s per pick**.

Dwell time = 1/3 × 0.30 s = **0.10 s**.

In that 0.10 s the shuttle must make its crossing through a stationary, fully open shed — which is why dwell, loom speed and shuttle speed can never be set independently of one another.

6. Capacity and limits of tappet shedding

- **Simple weaves only.** Plain weave, simple twills and other short, regular constructions whose frames follow a smooth, repeatable rise-and-fall are natural tappet work; irregular and figured designs are not.
- **Number of frames.** Each frame needs its own tappet, follower and linkage across a limited shaft width. In practice tappet shedding serves up to about **8–10 heald shafts**; beyond that a dobby gives more pattern power for the space.
- **Picks per repeat.** One revolution must cover one full repeat, divided around one circumference: more picks per repeat means less angular room for each lift and dwell, and a steeper, less practical profile. Repeats are therefore short — a few picks, not the dozens a dobby chain can store.
- **Pattern change is a mechanical change.** A new weave means new tappets and re-keying, with stoppage and skilled setting time. Tappet looms suit long runs of one simple cloth.

- **What tappets do well.** Robust, cheap to maintain and exact — a correctly cut tappet cannot misread a pick. For bulk plain shirtings, sheetings and simple twills, that reliability is why the tappet survived.

7. Tappet setting and timing

Setting a tappet system means establishing three things, in order:

1. **Correct tappets, correctly keyed.** Each tappet must be the profile for its frame's lift sequence, keyed at the angle that puts its lift on the right pick (half a revolution apart for the plain-weave pair, Figure 2). One tooth of keying error shifts a whole sequence onto the wrong picks.
2. **Correct lift.** Lever lengths are adjusted so heald travel equals the designed lift (Worked example 1) — enough for a clear shed, and no more. Check at the heald eye, frame by frame, equalised across the width.
3. **Correct timing.** Advance or retard the tappet shaft until shed change falls at its designed crank angles (Topic 3, Section 6): fully open before picking, dwell covering the shuttle's passage, no closing while it is inside. Set with the loom turned by hand to the marked angles.

Fault-finding follows the same divisions: a systematically wrong weave points to tappets or keying; a shed of wrong size points to lift, linkage or (on a negative system) tired return springs; a shed of right size at the wrong moment points to timing.

8. Tappet, dobby and jacquard — preview comparison

FEATURE	TAPPET	DOBBY	JACQUARD
Pattern stored in	The cam profiles themselves	Pegged lattice / chain, or electronic data read pick by pick	Cards or electronic data, one record per pick
Unit controlled	Heald frame, driven continuously by its cam	Heald frame, selected independently each pick	Individual ends or small groups (hooks and harness)
Practical frame / control capacity	About 8–10 frames	About 16–24 frames	Hundreds to thousands of hooks
Pick repeat	Short — a few picks per tappet revolution	Long — set by chain / data length	Very long — set by card / data length
Designs suited	Plain weave, simple twills and regular small	Medium patterns, extra-weft and multi-frame	Large figured designs

	constructions	weaves	
Cost of a pattern change	New tappets and re-setting — slow	New chain / program — moderate	New cards / data — design work, but no cam cutting

Topics 5 and 6 take up dobby and jacquard in full. Each step trades mechanical simplicity for pattern freedom; the tappet stands at the simple, dependable end.

Glossary

Tappet — a cam whose varying-radius profile drives one heald frame. **Throw** — maximum minus minimum radius; the displacement at the follower. **Follower** — the roller or arm riding on the profile. **Lamia (lamella)** — a flat connecting link between follower, treadle and heald frame. **Treadle** — the lever through which the frame is moved on many conventional looms. **Dwell** — the constant-radius portion of the profile, and the period the heald is held at full lift and the shed held open. **Bottom (under) shaft** — the shaft carrying the tappets, at half crank speed for a two-pick plain-weave repeat. **Negative tappet** — the tappet lifts, springs or weights return. **Positive tappet** — the mechanism drives the frame in both directions. **Lever ratio** — the ratio by which the linkage multiplies the throw.

Key points

• A tappet stores the weave in its profile: radius change lifts or lowers the follower, constant radius gives dwell, and a lever train converts throw into heald lift.
• Heald lift = throw × lever ratio — throw at the cam and lift at the heald eye are different numbers and must not be confused.
• For plain weave the tappet (bottom) shaft turns at half crank speed — one tappet revolution covers the two-pick repeat; in general, one revolution covers one full pattern repeat.
• Dwell of about one-third of the pick cycle at full lift gives the shuttle a stationary, fully open shed; at 200 picks/min that dwell is 0.10 s.
• Negative tappets lift positively but rely on springs or weights to return the frame; positive tappets drive the frame in both directions.
• Tappet shedding suits simple weaves only — practically up to about 8–10 frames with short pick repeats — and a pattern change means a mechanical change.

Practice

1. A tappet has maximum radius 44 mm and minimum radius 32 mm, driving its heald through a lever ratio of 1.8. Calculate the throw and the heald lift.

2. A loom runs at 180 picks per minute and its tappets give a dwell of 1/3 of the pick cycle at full lift. Calculate the time for one pick and the dwell time in seconds.
3. Explain, with the bottom-shaft gearing, why a plain-weave tappet shaft turns at half the crankshaft speed. What must one tappet revolution correspond to for a weave repeating on four picks?
4. Distinguish the lift portion and the dwell portion of a tappet profile, stating what the heald frame does while each passes under the follower.
5. A mill wants (a) a plain shirting in a very long run and (b) a 20-frame pattern repeating on 60 picks. State which case suits tappet shedding and why the other does not, referring to frame capacity and picks per repeat.
6. On a negative tappet loom the shed becomes progressively less clear over a shift, although the tappets and timing are unchanged. Give the most likely cause in the return system and how you would confirm it.

Answer key

1. Throw = 44 − 32 = **12 mm**. Heald lift = 12 × 1.8 = **21.6 mm**.
2. One pick = 60 ÷ 180 = **0.333 s**. Dwell = 1/3 × 0.333 s ≈ **0.111 s** (about 0.11 s).
3. Plain weave repeats every 2 picks and each frame must complete one lift-and-return cycle per repeat, so the tappet must complete one revolution in 2 picks: bottom-shaft speed = crank speed ÷ 2. For a four-pick repeat, one tappet revolution must correspond to **four picks** — the shaft turns once while the crankshaft turns four times.
4. Lift portion: radius changes between minimum and maximum, so the follower and heald frame travel — rising as radius grows, falling as it shrinks. Dwell portion: radius is constant (maximum for a raised frame, minimum for a lowered frame), so the follower does not move and the heald is held at full lift or fully lowered, keeping the shed fully open (or the frame fully down).
5. Case (a) suits tappet shedding: two frames, a two-pick repeat, and a long run over which the cost of cutting and setting tappets is spread. Case (b) does not: 20 frames exceeds the practical tappet capacity of about 8–10 frames, and 60 picks per repeat cannot be divided usefully around one tappet circumference — it needs a dobby (or, for figured work, a jacquard).
6. Most likely: the return springs (or reversing motion) on the negative system have weakened or one has broken, so frames no longer follow the falling side of the profile promptly and fully. Confirm by turning the loom slowly by hand and watching each frame return as its tappet radius falls, checking spring tension and condition frame by frame; a frame that lags by hand will lag worse at speed.

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